



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

AIOT

20-02-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	C	A	A	A	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	A	A	B	B	B	C	B	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	8.00	28.00	1.00	40.00	5.00	1.00	3.00	1.00	4.00	4.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	B	C	B	D	C	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	C	B	A	A	C	A	B	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	6.00	5.00	2.00	4.00	12.00	3.00	5.00	20.00	30.00	14.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	C	D	C	D	A	A	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	C	C	B	B	D	C	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	11.00	3.00	8.00	1.00	11.00	0.84	100.00	8.00	1.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (A)**

$$\vec{r} \cdot \vec{v} = -ive$$

2. **Ans (B)**

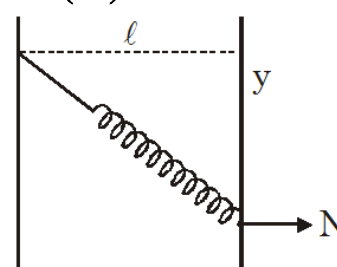
Let the tension in topmost string is T and acceleration of topmost mass is a.

$$\frac{T}{g} = \frac{T/2}{g-a}$$

$$2g - 2a = g$$

$$a = \frac{g}{2}$$

3. **Ans (D)**



$$N = k\sqrt{y^2 + \ell^2} \frac{\ell}{\sqrt{\ell^2 + y^2}} = k\ell$$

using work energy theorem

$$mgy - (k\ell)\mu y + \frac{1}{2}k\{\ell^2 - (\ell^2 + y^2)\} = 0$$

$$\mu k\ell y = mgy - \frac{1}{2}ky^2$$

$$ky = 2mg - 2\mu k\ell$$

$$y = \frac{2mg}{k} - 2\mu\ell$$

4. **Ans (C)**

For collision between M & m_1

$$V m_1 = \frac{2m_1}{m_1 + M} V$$

for collision between m_1 & m_2

$$V m_2 = \frac{2m_2}{(m_1 + m_2)(m_1 + M)} V$$

$$= \frac{4m_2 m_1 V}{m_1^2 + M m_1 + m_2 m_1 + m_2 M}$$

$$\frac{dV m_2}{dm_1} = 0 \Rightarrow 4m_2 V (m_1^2 + M m_1 + m_1 m_2 + m_2 M)$$

$$- 4m_2 m_1 V (2m_1 + M + m_2)$$

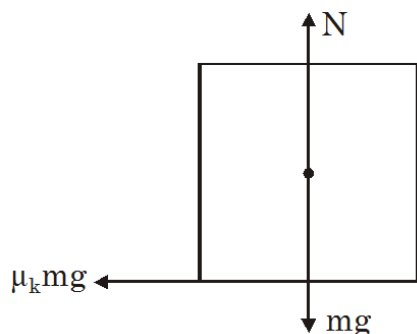
$$\Rightarrow m_2 m_1^2 + M m_1 m_2 + m_1 m_2^2 + m_2^2 M = 2m_1^2 m_2$$

$$- M m_1 m_2 - m_1 m_2^2 = 0$$

$$m_2^2 M = m_1^2 m_2 \Rightarrow m_1 = \sqrt{M m_2}$$

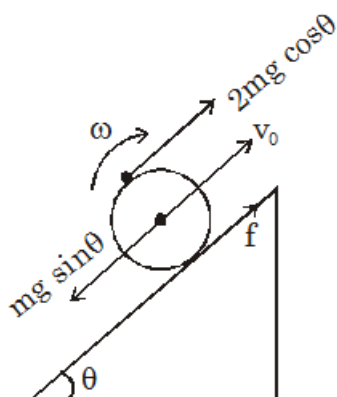
5. **Ans (A)**

Once slipping starts, torque of friction about COM will remain constant.



6. **Ans (A)**

As $2mg \cos \theta$ acts at the point of percussion of ring so it will not affect the friction so, direction will be decided by $mg \sin \theta$ only.



$$v = \omega R$$

As $mg \sin \theta$ decreases the velocity so for pure rolling ' ω ' should also decrease so friction will act upward the inclined plane whose torque decreases the ' ω '.

8. **Ans (A)**

Rate of loss of heat = $\frac{dQ}{dt} = \sigma e A (\theta^4 - \theta_s^4)$ is equal

$$\left(\frac{d\theta}{dt} \right) = \frac{\sigma e A}{ms} (\theta^4 - \theta_s^4)$$

Since mass of hollow sphere is less its $\frac{d\theta}{dt}$ is more

∴ Hollow sphere cools faster

9. **Ans (A)**

From first law of thermodynamics,

$$dQ = dU + dW$$

$$dQ = nC_v dT + PdV$$

$$\frac{dQ}{T} = nC_v \frac{dT}{T} + \frac{P dV}{T} \quad \dots(1)$$

The entropy change is

$$\Delta S = \int_i^f \frac{dQ}{T} \quad \dots(2)$$

$$= nC_v \int_i^f \frac{dT}{T} + \int_i^f \frac{P}{T} dV \quad \dots(3)$$

From ideal gas equation

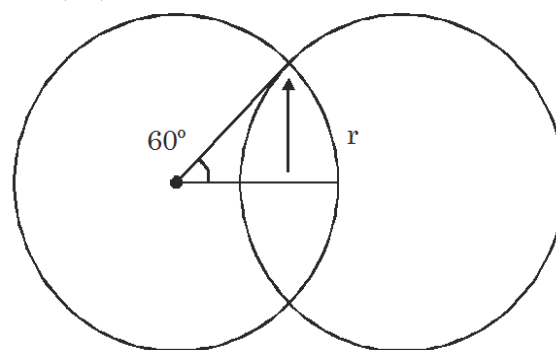
$$\frac{P}{T} = \frac{nR}{V}$$

Substituting this into second term of eq.(3), we have

$$\Delta S = nC_v \int_i^f \frac{dT}{T} + nR \int_i^f \frac{dV}{V}$$

$$= nC_v \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$$

10. **Ans (B)**



11. **Ans (C)**

$$\frac{KQ}{r} + \frac{KQ}{2r} + \frac{KQ}{4r} + \dots$$

$$\frac{KQ}{r} \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \right]$$

$$\frac{KQ}{r} \left[\frac{1}{1 - 1/2} \right] = \frac{2KQ}{r}$$

12. **Ans (A)**

at A, before orbit transfer

$$V_A = \sqrt{\frac{GM_e}{2R_e}}$$

Immediately after orbit transfer begins

$$E_{\text{total}} = -\frac{GM_em}{6R_e} = \frac{1}{2}m(V'_A)^2 - \frac{GM_em}{2R_e}$$

$$\frac{GM_e}{R_e^2} = g$$

$$\text{So } \Delta V = V'_A - V_A \simeq 875 \text{ m/s}$$

13. **Ans (A)**

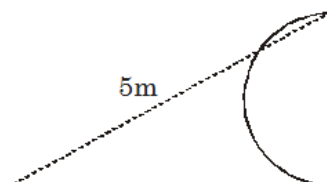
$$(20 + 10) \frac{1 \times 10^{-3}}{2} = s \times \left(10 - \frac{1 \times 10^{-3}}{2} \right)$$

$$\Rightarrow s \simeq 1.5 \text{ m}\Omega$$

14. **Ans (B)**

$$FB = i\ell B$$

$$= 2 \times 5 \times 2 = 20 \text{ N}$$



15. **Ans (B)**

Induced electric field is non conservative in nature.

17. **Ans (C)**

$$\frac{mv^2}{r} = \frac{kZe^2}{r^2} \Rightarrow \text{K.E.} \propto \frac{1}{r}, \text{ which increases,}$$

$$\text{but P.E.} = \frac{-kze^2}{r} \text{ decreases}$$

18. **Ans (B)**

Total energy of hydrogen atom in ground state
 $= -13.6 \text{ eV}$

$$\therefore M_H = M_P + M_e - \frac{13.6 \text{ eV}}{c^2}$$

19. **Ans (D)**

Image distance is 30 cm

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f}$$

$$\frac{1}{30} = \frac{2}{f}$$

$$f = 60 \text{ cm}$$

As optical axis shifts down by $(y - x)$ for second lens

$$m = \frac{y - x - 1}{y - x}$$

$$y = 4.5 \text{ cm}$$

20. **Ans (A)**

Reading of voltmeter A are closer, so it is more precise but its peak reading is away from actual reading. So it is less accurate as compared to B.

PART-1 : PHYSICS

SECTION-II

1. **Ans (8.00)**

$$\text{Magnetization} = \frac{\text{Magnetic moment}}{\text{volume}}$$

$$= \frac{9.27 \times 10^{-24} \times \frac{\text{Volume} \times \text{density}}{\text{Molar mass}} \times N_a}{\text{Volume}}$$

$$= \frac{9.27 \times 10^{-24} \times 8000 \times 6.023 \times 10^{23}}{56 \times 10^{-3}}$$

$$= 8 \times 10^5 \text{ A-m}$$

2. **Ans (28.00)**

$$\omega_D = \frac{2\pi}{T_D}$$

$$\omega_D^2 = \omega_N^2 - \gamma^2$$

Where γ damping coefficient

$$\omega_D = 24$$

$$\omega_N = \sqrt{\frac{k}{m}} = \sqrt{625}$$

$$\therefore \gamma = 7$$

$$\gamma \text{ (Damping coefficient)} = \frac{b}{2m}$$

$$(b) \text{ damping constant} = 28$$

3. **Ans (1.00)**

$$f_0 + f_e = 100$$

$$\frac{f_0}{f_e} = 9$$

$$f_e = 10$$

$$f_0 = 90$$

$$v = 90$$

$$f_e = 10$$

$$-\frac{1}{90} - \frac{1}{u} = \frac{1}{10}$$

$$\frac{1}{u} = -\frac{1}{90} - \frac{1}{10} = \frac{-10}{90}$$

$$u = -9 \text{ cm}$$

$$\Rightarrow \Delta x = 1 \text{ cm}$$

4. **Ans (40.00)**



$$f = f_0 \left(\frac{v + v_0}{v + v_s} \right)$$

$$\Rightarrow 760 = 720 \left(\frac{340 + v_0}{340 + 20} \right)$$

$$\Rightarrow \frac{760}{720} = \frac{340 + v_0}{360}$$

$$340 + v_0 = \frac{760}{720} \times 360$$

$$= 380$$

$$v_0 = 380 - 340$$

$$v_0 = 40 \text{ m/s}$$

5. **Ans (5.00)**

From graph 1

$$\lambda = 4 \text{ cm}$$

from graph 2

$$T = 0.8 \text{ sec}$$

$$\& \text{ speed} = \frac{\lambda}{T} = \frac{4}{8 \times 10^{-1}} = 5 \text{ cms}^{-1}$$

6. **Ans (1.00)**

$$\alpha = \frac{\lambda}{d} \quad \alpha \text{ in water} = 0.2^\circ$$

$$\lambda_{\text{water}} = 6000 \text{ \AA}$$

$$\lambda_{\text{air}} = \frac{4}{3} \times 6000 \left[\mu = \frac{\lambda_{\text{air}}}{\lambda_{\text{medium}}} \right]$$

$$= 8000 \text{ \AA}$$

$$\lambda_{\text{liq.}} = \frac{8000}{\frac{8}{5}} = 5000 \text{ \AA}$$

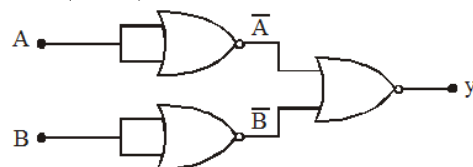
$$= A s \propto \lambda$$

$$\frac{\alpha_{\text{water}}}{\alpha_{\text{liquid}}} = \frac{\lambda_{\text{water}}}{\lambda_{\text{liquid}}}$$

$$= \frac{0.2^\circ}{\alpha_{\text{liquid}}} = \frac{6000}{5000}$$

$$\alpha_{\text{liquid}} = \frac{5000 \times 0.2}{6000} = \frac{1^\circ}{6}$$

7. **Ans (3.00)**



8. **Ans (1.00)**

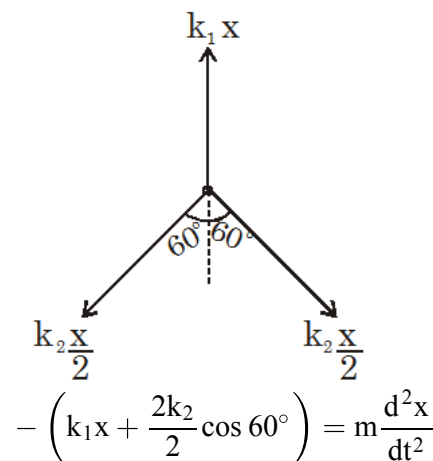
$$f_a = \frac{1}{2\pi} \sqrt{\frac{\left(k_1 + \frac{k_2}{2}\right)}{m}},$$

$$f_b = \frac{1}{2\pi} \sqrt{\frac{\left(k_1 + \frac{k_2}{2}\right)}{m}}$$

$$(a) \text{ In case (a) } k_{\text{eq}} = k_1 + \frac{k_2}{2}$$

$$(b) x' = x \cos 60^\circ$$

Thus making FBD



$$\text{In this case also } k_{\text{eq}} = k_1 + \frac{k_2}{2}$$

$$\text{So } f_b = f_a$$

9. **Ans (4.00)**

Heat lost = Heat gained

Let m_s = mass of steam

$$m_s \times 540 + m_s \times 1 \times 10 = 2 \times 10^3 \times 1 \times 55$$

$$m_s 550 = 2 \times 10^3 \times 55$$

$$m_s = 2 \times 10^2 \text{ gm}$$

$$\alpha = 2, \beta = 2$$

$$\alpha + \beta = 4$$

10. **Ans (4.00)**

$$V_R = 3V$$

$$V_{2R} = 6V$$

$$C(V_0 - 6) + 2C(V_0 - 9) + 3C(V_0 - 0) = 0$$

$$V_0 = 4V$$

$$\therefore Q_C = 4 \mu C$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (C)**

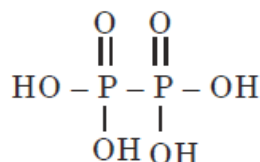
A' is Ne $\Rightarrow \Delta_{eg}H = 116 \text{ kJ/mole}$

'B' is Cl $\Rightarrow \Delta_{eg}H = -349 \text{ kJ/mole}$

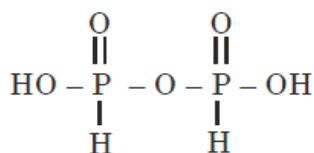
'C' is Se $\Rightarrow EA : S > Se > Te > O$

'D' is Sb $\Rightarrow EA : P > As > Sb > N$

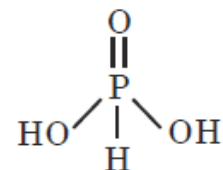
2. **Ans (D)**



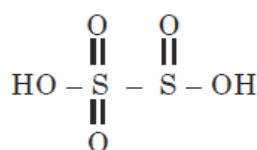
Hypophosphoric acid



pyrophosphorous acid

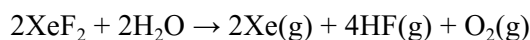
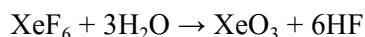
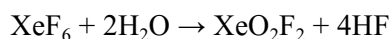
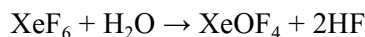


phosphorous acid



pyrosulphurous acid

3. **Ans (C)**



5. **Ans (C)**

Wrought iron is purest form of commercial iron.

Zinc is used for galvanising iron

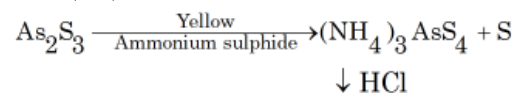
Distillation is used for Hg

6. **Ans (B)**

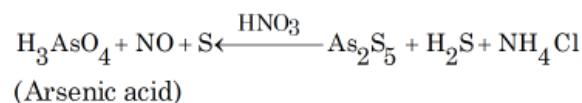
Energy released on combustion :

Energy (in kJ)	H ₂ (g)	H ₂ (l)	CH ₄ (g)
per mole	286	285	880
per g	143	142	53
per L	12	9968	35

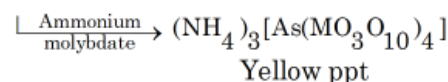
7. **Ans (D)**



↓ HCl

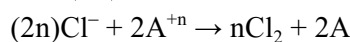


(Arsenic acid)



Yellow ppt

8. **Ans (C)**



Applying Nerst equation

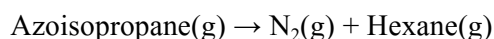
$$0.02 = 0.04 - \frac{0.06}{2n} \log \left(\frac{1}{10^{-2}} \right)$$

$$0.02 = 0.04 - \frac{0.06}{2n} \times 2$$

$$-0.02 = -\frac{0.06}{n}$$

$$n = 3$$

10. **Ans (A)**



$$K = \frac{2.3}{230} \log \frac{180}{45}$$

$$K = \frac{1}{100} \log 4$$

$$K = 6 \times 10^{-3} \text{ sec}^{-1}$$

11. **Ans (D)**

Factual

12. **Ans (C)**

Statement A : ΔG° may or may not be equal to zero

Statement B : Gibbs function attains minimum value.

Statement D : $(\Delta G)_{T,P} < 0$

13. **Ans (C)**

Let salt be A_xB_y

$$K_{sp} = (x \times s)^x (y \times s)^y$$

$$\left(\frac{K_{sp}}{x^x \times y^y} \right) = s^{x+y}$$

$$\log(s) = \left(\frac{1}{x+y} \right) \left[\log \frac{K_{sp}}{x^x y^y} \right]$$

$$\log = \frac{1}{x+y} [\log K_{sp} - [x \log x + y \log y]]$$

Solve for 'x & y' by putting value of x and y.

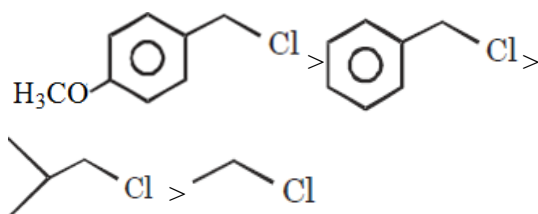
for x = 1 and y = 2

$$\log(s) = \frac{1}{3} [-5 - [0.6]]$$

$$\log(s) = -\frac{56}{30}$$

14. **Ans (B)**

Correct order,



15. **Ans (A)**

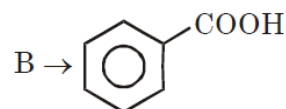
A : Saccharic acid $\rightarrow$ No. of stereoisomers = 10

B : Gluconic acid $\rightarrow$ No. of enantiomeric pair = 8

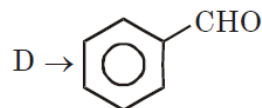
C : n-hexane $\rightarrow$ No. of structural isomers = 5

D : 5 moles of acid is obtained

16. **Ans (A)**



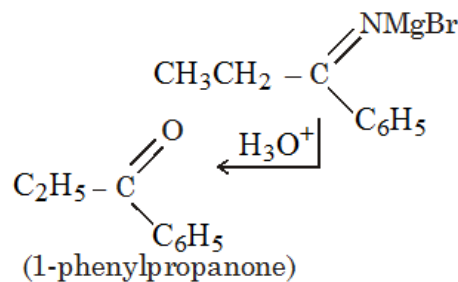
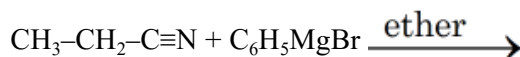
C $\rightarrow$ CHI_3



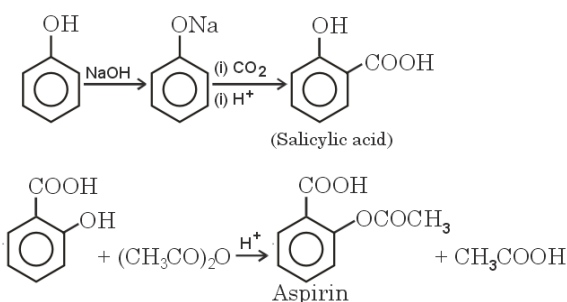
17. **Ans (C)**

Gabriel synthesis is not used for preparation of aromatic amines

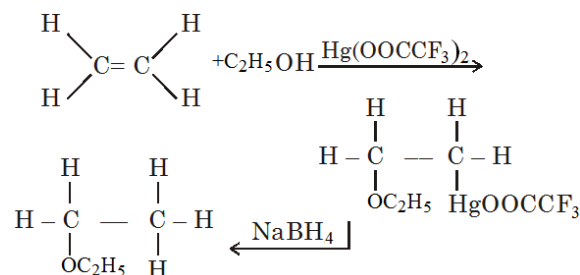
18. **Ans (A)**



19. **Ans (B)**



20. **Ans (D)**



PART-2 : CHEMISTRY

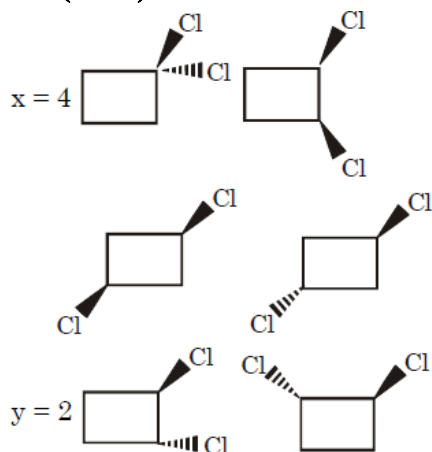
SECTION-II

2. **Ans (5.00)**

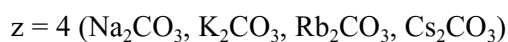
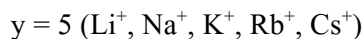
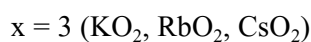
X = 3

Y = 1

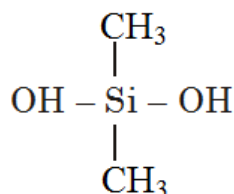
3. Ans (2.00)



4. Ans (4.00)



5. Ans (12.00)



6. Ans (3.00)

(b, c, d)

7. Ans (5.00)

$$(\text{P.E.})_{n=2} = -\frac{40}{(2)^2}$$

$$= -10 \text{ eV}$$

$$(\text{K.E.})_{n=2} = 5 \text{ eV}$$

8. Ans (20.00)

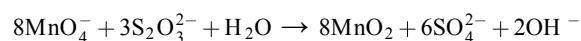
$$PV = nRT$$

$$PT = \left(\frac{nR}{V} \right) T^2$$

$$y = cx^2 \quad \frac{\partial y}{\partial x} = (2C)x$$

$$2 \times \frac{nRT}{V} = 60$$

9. Ans (30.00)



$$0.4 \times 3 \times 40 = 0.2 \times 8 \times V$$

$$\Rightarrow V = \frac{48}{1.6} = 30 \text{ ml}$$

10. Ans (14.00)

$$\Delta G = nRT \ln \left(\frac{P_2}{P_1} \right)$$

$$p \ln 2 = 20 \times 0.7 = 14 \text{ Latm}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (D)

$$|A^T \cdot \text{adj}(\text{adj}A)| = |A| \cdot |A|^{2^2} = |A|^5$$

$$\Rightarrow |A|^5 = 3^5 \Rightarrow |A| = 3$$

$$|A|^3 = |A^3| = 27$$

$$5p^2 - 5 + 12 = 27$$

$$p^2 = 4 \Rightarrow p = \pm 2$$

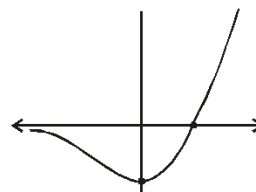
2. Ans (A)

$$f'(x) = \frac{e^x \cdot (x^3 - 3x^2 + \frac{9x}{2})}{(x^2 + x + \frac{3}{2})^2} = 0$$

$$\Rightarrow x = 0$$

$$\frac{-}{+}$$

$$x = 0$$



function is $\uparrow$ in $(0, \infty)$

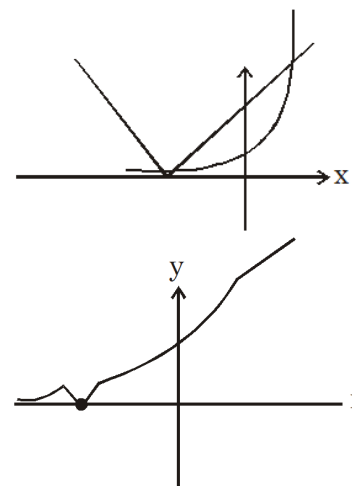
& function is $\downarrow$ in $(-\infty, 0)$

3. Ans (C)

$$\frac{1}{\alpha\beta\gamma} \cdot \left[\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \right] \Rightarrow \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{(\alpha\beta\gamma)^2} = \frac{4}{4} = 1$$

4. Ans (D)

Graph of $f(x)$



5. **Ans (C)**

at $\lambda = 5$ first and second lines are co-incident

6. **Ans (D)**

$(a, a) \in R$ since $a = 3^0 \cdot a$, R is reflexive

if $(a, b) \in R \Rightarrow a = 3^k \cdot b, k \in I$

$\Rightarrow b = 3^{-k} \cdot a, -k \in I \Rightarrow (b, a) \in R$

R is symmetric

if $(a, b) \& (b, c) \in R$

$a = 3^{k_1} \cdot b, b = 3^{k_2} \cdot c; k_1, k_2 \in I$

$\Rightarrow a = 3^{k_1+k_2} \cdot c; (k_1 + k_2) \in I$

$(a, c) \in R$ R is transitive

$\therefore R$ is an equivalence relation.

7. **Ans (A)**

$$\frac{\frac{b}{a}}{1 + \frac{b}{a}} = 3b \Rightarrow a + b = \frac{1}{3}$$

use A.M > G.M.

$$\frac{a+b}{2} > (ab)^{1/2}$$

$$(ab)^{1/2} < \frac{1}{6}$$

$$ab < \frac{1}{36}$$

8. **Ans (A)**

lim is 1^∞ form (only R.H.L. exist)

$$L = e^{\lim_{x \rightarrow 2} \left(\frac{1}{\sqrt{x-2}} \right) \cdot \sqrt{\tan(x-1)(x-2)}}$$

$$L = e^{\lim_{x \rightarrow 2} \left(\sqrt{\frac{\tan(x-1)(x-2)}{(x-1)(x-2)}} \right) \cdot \sqrt{x-1}}$$

$$L = e^1$$

9. **Ans (C)**

$$\text{put } \frac{n}{3} = k$$

$$L = \lim_{k \rightarrow \infty} \left(\frac{(k+1)(k+2)\dots(3k)}{k^{2k}} \right)^{1/k}$$

$\Rightarrow$ taking log

$$\ln L$$

$$\Rightarrow \lim_{k \rightarrow \infty} \frac{1}{k} \left[\ln \left(1 + \frac{1}{k} \right) + \ln \left(1 + \frac{2}{k} \right) + \dots + \ln \left(1 + \frac{2k}{k} \right) \right]$$

$$\Rightarrow \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{r=1}^{2k} \left(\ln \left(1 + \frac{r}{k} \right) \right)$$

$$\Rightarrow \int_0^2 \ln(1+x) \Rightarrow \left(x \ln(1+x) - \int \frac{x}{x+1} dx \right)_0^2$$

$$\Rightarrow [x \ln(1+x) - x + \ln(1+x)]_0^2$$

$$\ln L \Rightarrow (2 \ln 3 - 2 + \ln 3) \Rightarrow 3 \ln 3 - 2$$

$$L = e^{3 \ln 3 - 2} \Rightarrow \frac{27}{e^2}$$

10. **Ans (A)**

Take x^{15} common from D^r

$$I = \int \frac{(1-2x)dx}{(1-x^2+x)^{\frac{1}{3}}}$$

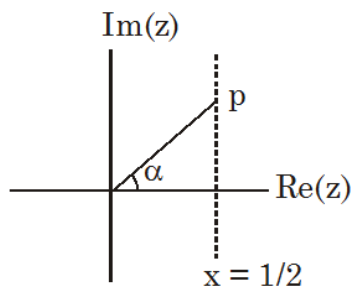
put $1-x^2+x = t$

$$\int \frac{dt}{(t)^{\frac{1}{3}}} = \frac{3}{2} t^{\frac{2}{3}} + C = \frac{3}{2} (1-x^2+x)^{\frac{2}{3}} + C$$

11. **Ans (B)**

Let $z = e^{i\theta} = ; \theta \neq 2n\pi$

$$\begin{aligned}\frac{1}{1-z} &= \frac{1}{1-\cos\theta - i\sin\theta} \\ &= \frac{1}{2\sin\frac{\theta}{2}\left(\sin\frac{\theta}{2} - i\cos\frac{\theta}{2}\right)} \\ &\Rightarrow \frac{1}{-2\sin\frac{\theta}{2} \cdot i \cdot e^{i\theta/2}} \Rightarrow \frac{\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}}{-2i\sin\frac{\theta}{2}} \\ &\Rightarrow \frac{1}{2} + \frac{i\cot\frac{\theta}{2}}{2}\end{aligned}$$



so locus of $\frac{1}{1-z}$ is line $x = \frac{1}{2}$

Hence $\arg\left(\frac{1}{1-z}\right) = \alpha$

12. **Ans (C)**

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

so $f(x) = 0$

13. **Ans (C)**

$$P(X=1) = {}^5C_1 p \cdot q^4 = 0.4$$

$$P(X=2) = {}^5C_2 p^2 \cdot q^3 = 0.6$$

$$\frac{{}^5C_1 \cdot p q^4}{{}^5C_2 p^2 q^3} = \frac{2}{3} \Rightarrow \left(\frac{5}{10}\right) \left(\frac{q}{p}\right) = \frac{2}{3}$$

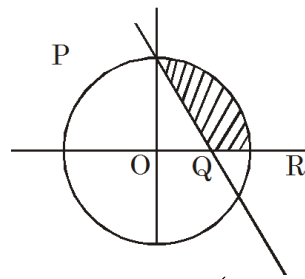
$$p = \frac{3q}{4} \quad \dots(1)$$

$$p + q = 1 \quad \dots(2)$$

$$p + \frac{4p}{3} = 1 \Rightarrow p = \frac{3}{7}; q = \frac{4}{7}$$

$$P(X=3) = {}^5C_3 p^3 \cdot q^2 = 10 \times \left(\frac{3}{7}\right)^3 \left(\frac{4}{7}\right)^2 = \frac{4320}{7^5}$$

14. **Ans (C)**



$$\begin{aligned}\text{required area is } &\left(\frac{\pi}{4} \cdot 5 - \frac{1}{2} \cdot \sqrt{5} \cdot 1\right) \\ &\Rightarrow \frac{5\pi - 2\sqrt{5}}{4}\end{aligned}$$

15. **Ans (B)**

Grouping

$$9 \rightarrow 6, 2, 1$$

$$9 \rightarrow 5, 3, 1$$

$$9 \rightarrow 4, 3, 2$$

$$\begin{aligned}\text{so } &\left(\frac{9!}{6!2!1!} + \frac{9!}{5!3!1!} + \frac{9!}{4!3!2!}\right) \times 3! \\ &= \left(\frac{9 \times 8 \times 7}{2} + \frac{9 \times 8 \times 7 \times 6}{6} + \frac{9 \times 8 \times 7 \times 6 \times 5}{6 \times 2}\right) \times 3! \\ &= (9 \times 4 \times 7 + 9 \times 8 \times 7 + 9 \times 4 \times 7 \times 5) \times 6 \\ &= 2016 \times 6\end{aligned}$$

16. **Ans (B)**

$$\frac{dy}{dx} = \frac{(x-3)^2 + (y-2)^2}{(x-3)(y-2)}$$

put $x-3 = X$ and $y-2 = Y$

$$\frac{dY}{dX} = \frac{X^2 + Y^2}{XY}$$

now $Y = vX$

$$\Rightarrow v + x \frac{dv}{dX} = \frac{v^2 + 1}{v} = v + \frac{1}{v}$$

$$X \frac{dv}{dX} = \frac{1}{v}$$

$$\frac{dX}{X} = v dv$$

$$\ln X = \frac{v^2}{2} + c$$

$$\Rightarrow \ln(x-3)^2 = \left(\frac{y-2}{x-3}\right)^2 + c$$

17. Ans (D)

$$16^{\left|1+\alpha+\alpha^2+\frac{1}{\alpha^2}-\frac{1}{\alpha}\right|} = 16^{\left|\frac{\alpha^2+\alpha^3+\alpha^4+1-\alpha}{\alpha^2}\right|}$$

$$16^{\left|\frac{1+\alpha+\alpha^2+\alpha^3+\alpha^4-2\alpha}{\alpha^2}\right|} = 16^{\left|\frac{-2\alpha}{\alpha^2}\right|}$$

$$= 16^2 = 256$$

$$\therefore |\alpha| = 1$$

18. Ans (C)

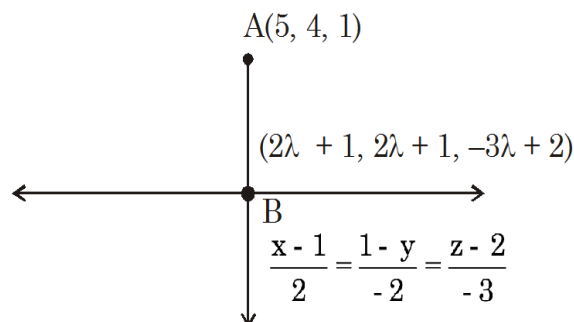
$$s = 1 + 2\tan x \cdot \tan y + 3(\tan x \cdot \tan y)^2 + \dots \infty, (\text{A.G.P.})$$

$$\frac{s(\tan x \cdot \tan y) = \tan x \cdot \tan y + 2(\tan x \cdot \tan y)^2 + \dots \infty}{s(1 - \tan x \cdot \tan y) = 1 + \tan x \cdot \tan y + (\tan x \cdot \tan y)^2 + \dots \infty}$$

$$s = \frac{1}{(1 - \tan x \cdot \tan y)^2}$$

19. Ans (C)

Here lines are perpendicular



$$\vec{AB} = (2\lambda - 4, 2\lambda - 3, -3\lambda + 1)$$

$$\vec{AB} \cdot (2, 2, -3) = 0 \Rightarrow \lambda = 1$$

$$B = (3, 3, -1)$$

$$\text{mid point of AB, is } M = \left(4, \frac{7}{2}, 0\right)$$

then distance of point of M from the point P

$$\left(-2, \frac{3}{2}, 3\right) \text{ is } 7$$

20. Ans (C)

$${}^{100}C_{42} = {}^{100}C_{58} \text{ and } {}^{100}C_{46} = {}^{100}C_{54}$$

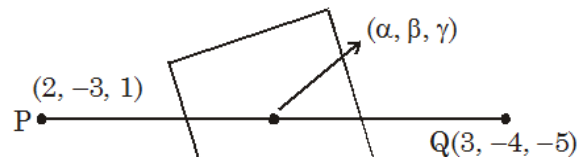
$$\Rightarrow {}^{100}C_{42} \cdot {}^{100}C_{46} - {}^{100}C_{54} \cdot {}^{100}C_{58} = 0$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (11.00)

Let α, β and γ are roots of $ax^3 + bx^2 + cx + d = 0$



Plane

$$\text{Let line PQ : } \frac{x-2}{1} = \frac{y+3}{-1} = \frac{z-1}{-6} = \lambda$$

any point on line $(\lambda + 2, -\lambda - 3, -6\lambda + 1)$

also lies on given plane $2x + y + z = 7$

$$\Rightarrow 2(\lambda + 2) - \lambda - 3 - 6\lambda + 1 = 7$$

$$-5\lambda + 2 = 7 \Rightarrow \lambda = -1$$

$$\therefore \text{hence point } (\alpha, \beta, \gamma) = (1, -2, 7)$$

$$\alpha + \beta + \gamma = 1 - 2 + 7 = 6 \Rightarrow -b/a = 6 \dots (1)$$

$$\Sigma \alpha\beta = -2 - 14 + 7 \Rightarrow \frac{c}{a} = -9 \dots (2)$$

$$\alpha\beta\gamma = (1)(-2)(7) \Rightarrow \frac{d}{a} = -14 \dots (3)$$

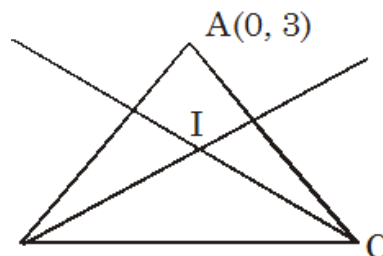
$$\text{hence } \frac{-b+c-d}{a} \text{ put values of (1), (2), (3)}$$

$$= 6 - 9 + 14$$

$$= 11$$

2. Ans (3.00)

Image of A in angle bisectors from B and C lies on line BC.



line BC is $y = 0$

image of A(0, 3) in $y = 0$ is (0, -3)

$$h - k = 3$$

3. **Ans (8.00)**

Equation of line passing through $P(1, 0, -3)$

and parallel to $\frac{x-2}{2} = \frac{y+2}{3} = \frac{z-6}{-6}$ is

$$\frac{x-1}{2} = \frac{y}{3} = \frac{z+3}{-6} = \lambda$$

co-ordinates of any point on the line are

$Q = (2\lambda + 1, 3\lambda, -6\lambda - 3)$ so, Q lies on plane

$$\Rightarrow (2\lambda + 1 - 3\lambda + 6\lambda + 3 = k)$$

$$\Rightarrow \lambda = \frac{k-4}{5}$$

so point Q is $\left(\frac{2k-3}{5}, \frac{3k-12}{5}, \frac{-6k+9}{5}\right)$

$$PQ = 7$$

$$\Rightarrow PQ = \sqrt{\left(\frac{2k-8}{5}\right)^2 + \left(\frac{3k-12}{5}\right)^2 + \left(\frac{-6k+24}{5}\right)^2} = 7$$

$$\Rightarrow 49k^2 - 392k + 441 = 0$$

4. **Ans (1.00)**

$$\sin(\cos^{-1}x) \Rightarrow \sin\left(\frac{\pi}{2} - \sin^{-1}x\right) = \cos(\sin^{-1}x)$$

$$(\cos(\sin^{-1}x)) = \cos\left(\frac{\pi}{2} - \cos^{-1}x\right) = \sin(\cos^{-1}x)$$

$$\Rightarrow \cos^{-1}(\cos(\sin^{-1}x)) = \sin^{-1}x \text{ for } x \in (0, 1)$$

$$\Rightarrow \sin^{-1}(\cos(\sin^{-1}x)) = \sin^{-1}(\sin(\cos^{-1}x)) = \cos^{-1}x$$

$$\Rightarrow \tan\left(\frac{\pi/2}{2}\right) = 1$$

5. **Ans (11.00)**

$$P(1) = 2 \times 64 + 3^4 = 209$$

$$P(2) = 2 \times 4^5 + 3^7 = 4235$$

HCF of 209 and 4235 is 11.

Now to check if it is divisible by 11 $\forall n \in \mathbb{N}$.

Let $P(k) = 2 \cdot 4^{2k+1} + 3^{3k+1}$ is divisible by 11.

$$P(k) = 8 \cdot 16^k + 3 \cdot 27^k$$

$$P(k+1) = 2 \cdot 4^{2k+3} + 3^{3k+4}$$

$$= 8 \cdot 16^{k+1} + 3 \cdot 27^{k+1}$$

$$= 128 \times 16^k + 81 \cdot 27^k$$

$$= p(k) + 6 [20 \cdot 16^k + 13 \cdot 27^k]$$

$$= p(k) + 6 \cdot 33 \cdot 16^k - 6[13(27^k - 16^k)]$$

All terms are divisible by 11 as $x^n - y^n$ is

always divisible by $x-y$.

6. **Ans (0.84)**

$$P(E_1) = 0.5 \Rightarrow P(E'_1) = 0.5$$

$$P(E_2) = 0.6 \Rightarrow P(E'_2) = 0.4$$

$$P(E_3) = 0.2 \Rightarrow P(E'_3) = 0.8$$

$$P(E) = 1 - P(\text{no shot hit the enemy})$$

$$= 1 - 0.5 \times 0.4 \times 0.8$$

$$= 1 - 0.16 = 0.84$$

7. **Ans (100.00)**

$$\tan\theta = 2 - \sqrt{3} \text{ or } \tan\theta = 2 + \sqrt{3}$$

$$\text{for } \tan\theta = 2 - \sqrt{3}$$

number of solution = 50

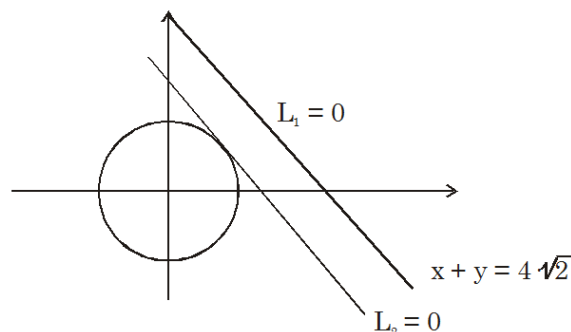
$$\text{for } \tan\theta = 2 + \sqrt{3}$$

number of solution = 50

total solution = 50 + 50

8. **Ans (8.00)**

$$x + y = \lambda$$



Distance between L_1 & L_2 is 2

$$\frac{|\lambda - 4\sqrt{2}|}{\sqrt{2}} = 2 \Rightarrow \lambda = 6\sqrt{2} \text{ or } 2\sqrt{2}$$

equation of L_2 is

$$x + y = 2\sqrt{2}$$

9. Ans (1.00)

- (i) $p \vee q$ is T, $\sim p \wedge q$ is F $\Rightarrow$ (i) is F
 (ii) $p \wedge \sim q$ is T, $\sim p \vee q$ is F $\Rightarrow$ (ii) is F
 (iii) $p \wedge q$ is F, $p \vee q$ is T $\Rightarrow$ (iii) is T
 (iv) $p \rightarrow q$ is F, $p \wedge q$ is F $\Rightarrow$ (iv) is F

10. Ans (4.00)

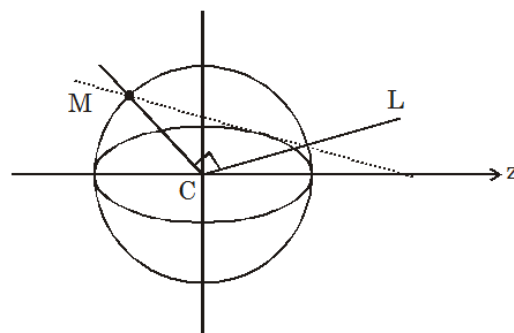
equation of tangent is

$$\frac{x}{a} \left(\frac{1}{2} \right) + \frac{y}{b} \left(\frac{\sqrt{3}}{2} \right) = 1 \quad \dots\dots(i)$$

$$\text{auxiliary circle is } x^2 + y^2 = a^2 \quad \dots\dots(ii)$$

combined equation of CL & CM is obtained by homogenising

(ii) with (i)



$$x^2 + y^2 - a^2 \left(\frac{x}{2a} + \frac{y\sqrt{3}}{2b} \right)^2 = 0$$

$$\angle LCM = 90^\circ$$

$$\Rightarrow 1 - \frac{1}{4} + 1 - \frac{3a^2}{4b^2} = 0$$

$$\boxed{\frac{a^2}{b^2} \Rightarrow \frac{7}{3}}$$

$$e = \sqrt{1 - \frac{b^2}{a^2}} \Rightarrow \frac{2}{\sqrt{7}}$$